

Evaluating the Generalization Limits of CRNN-CTC Text Recognition on Synthetic Documents: Font Shift, Degradation Robustness, and a Comparison with Character Detection

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Abstract — Convolutional Recurrent Neural Networks (CRNNs) supervised by Connectionist Temporal Classification (CTC) are widely deployed for sequence recognition, frequently relying on synthetic data to circumvent manual annotation costs. In this study, we empirically evaluate a CRNN-CTC model trained exclusively on synthetic document text lines. The experimental suite comprises 4,900 unique synthetic page layouts (47,533 unique underlying line instances) yielding 55,130 evaluated line instances across training, validation, and five controlled test conditions across 73 typographic classes plus whitespace. We analyze 12-epoch training dynamics, quantify the unseen font generalization penalty, evaluate robustness across controlled degradation severity tiers (Clean, Moderate, Hard), and compare 1D line transduction against a 2D YOLO11 character detector evaluated on identical document layouts.

Our verified findings demonstrate: (1) Training undergoes an observed initial output collapse in Epoch 1 where all predictions emit empty strings (100% Strict CER, 289,413 deletions) before gradient breakout in Epoch 2 (4.16% CER), converging to 1.79% validation CER at Epoch 12, an empirical trajectory hypothesized to reflect initial blank-token dominance under CTC alignment optimization; (2) In-distribution recognition on seen fonts under moderate degradation achieves 0.31% Strict CER and 89.91% exact line accuracy, whereas evaluating on held-out font families results in an error ratio of $12.87 \times$ (3.99% Strict CER, an increase of 3.67 percentage points) and a 24.2-fold explosion in character deletions (from 86 to 2,082); (3) Lookalike character normalization resolves 36.54% of errors on seen fonts, but only 5.04% on unseen fonts, confirming a shift from optical homoglyphs ($\frac{l}{i}, \frac{o}{0}$) to structural glyph misclassifications ($v \rightarrow u, b \rightarrow l$) and kerning space insertions; (4) Across identical page layouts, photometric degradation from Clean to Hard increases CER from 0.27% to 0.63% (a $2.33 \times$ ratio, +0.35 percentage points), showing that error rates under severe synthetic image noise remain lower than those observed under held-out font shifts; and (5) Under the evaluated implementation, high character-detection accuracy (YOLO11 mAP50 of 94.56%, mAP50–95 of 79.02%) did not translate into accurate reconstructed text transcription (42.86% CER, WER > 109%), illustrating the algorithmic distinction between spatial character localization and sequence transcription without implying universal superiority of either model family.

Keywords: Optical Character Recognition, CRNN, Connectionist Temporal Classification, Synthetic Data, Font Shift, Degradation Robustness, Character Detection.

1. Introduction

Automated text transcription from image representations is a central problem in document analysis. Early optical character recognition (OCR) systems relied on cascaded pipelines: binarization, line slicing, explicit glyph segmentation, and heuristic classification [6]. These cascaded architectures suffered from error propagation: any failure in character boundary segmentation corrupted subsequent classification.

The Convolutional Recurrent Neural Network (CRNN) proposed by Shi et al. [1] framed text recognition as an alignment-free sequence transduction task. By combining a Convolutional Neural Network (CNN) for translation-invariant visual feature extraction, a Bidirectional Long Short-Term Memory (BiLSTM) network for sequential context modeling, and Connectionist Temporal Classification (CTC) loss [2] for alignment-free supervision, the architecture eliminated explicit character segmentation.

To train deep sequence models without expensive manual annotation, synthetic document generation is standard practice [3, 4, 5]. Synthetic pipelines provide exact ground-truth annotations and parameterized control over image distortions (such as skew, blur, noise, and paper textures). However, synthetic training introduces an important question regarding generalization:

To what degree does a CRNN trained on synthetic document lines achieve font invariance versus visual overfitting to training font families, and how does font shift compare with image degradation?

This study evaluates this question empirically using a controlled synthetic document benchmark.

2. Related Work

2.1 CRNN and Sequence Recognition

Shi et al. [1] demonstrated that sequence-to-sequence learning on line strips significantly outperforms segmentation pipelines. CTC loss, introduced by Graves et al. [2], enables models to output label sequences

from unsegmented frame sequences by introducing a blank token ($\emptyset$) and marginalizing over all valid alignments.

2.2 Synthetic Text Generation

Jaderberg et al. [3, 4] and Gupta et al. [5] showed that synthetic engines can train robust text recognizers. In document OCR, synthetic engines enable controlled testing across degradation levels. However, synthetic models may exhibit performance drops when encountering unseen typefaces or real scanning artifacts.

2.3 Character Detection vs. Sequence Modeling

Object detection models, such as Ultralytics YOLO11 [7], detect small objects with high mean Average Precision (mAP). Applying 2D detectors to document character localization offers an alternative to 1D CRNN line transduction. However, converting 2D boxes into text lines requires reading-order reconstruction and space inferencing, presenting distinct algorithmic trade-offs.

3. Research Questions and Contributions

This study investigates six specific research questions:

- **RQ1 (In-Distribution Performance):** What transcription accuracy does the CRNN achieve on seen fonts under moderate degradation?
- **RQ2 (Degradation Robustness):** How does accuracy change across controlled degradation tiers (Clean, Moderate, Hard)?
- **RQ3 (Font Generalization Gap):** What is the quantitative performance drop on strictly held-out font families?
- **RQ4 (Error Patterns):** Which Levenshtein [8] edit operations (substitutions, deletions, insertions) dominate under visual shifts?
- **RQ5 (Homoglyph Disambiguation):** What fraction of errors are recoverable through visual lookalike normalization?
- **RQ6 (Detection vs. Transduction):** How does full-page YOLO11 character detection compare with line-level CRNN sequence recognition under the evaluated pipeline?

Contributions

1. We quantify the font generalization penalty of a synthetic CRNN model, measuring a $12.87\times$ CER ratio between held-out and seen font conditions (+3.67 percentage points).
2. Across identical page layouts, we document that severe photometric noise results in a $2.33\times$ CER ratio (+0.35 percentage points), showing that error rates under high synthetic noise remain below those on held-out fonts.
3. We examine the task formulation trade-off between 2D character detection and 1D sequence transduction under the evaluated implementations.

4. Materials and Methods

4.1 Dataset Generation Pipeline

Synthetic pages are produced using `synth_pages` at 1024 px width with variable heights (160 to 1024 px). Text lines are formed from random Latin word sequences sampled across a 73-character non-space vocabulary: a-z (26), A-Z (26), 0-9 (10), and punctuation . , ; : ! ? ' " - () (11). Rectified text line crops are exported at a fixed height of 48 px with preserved aspect ratios.

4.2 Font Segregation Protocol

Font families are partitioned using `split_fonts.py`:

- **Training Pool (train/)**: ≥ 40 font families used for model training and seen-font evaluation.
- **Validation Pool (val/)**: ≥ 8 font families reserved for epoch checkpointing.
- **Unseen Test Pool (unseen_test/)**: ≥ 10 font families held out entirely for testing.

4.3 Degradation Model and Layout Reuse

A parameter $s \in [0, 1]$ controls distortion severity (affine skew, stretch, warp, blur, noise, stains, and JPEG compression). Crucially, the three `test_tiers_*` partitions reuse the **same 300 page layouts and 3,800 underlying line instances**, rendering identical content under Clean ($s = 0.0$), Moderate ($s = 0.25$ – 0.50), and Hard ($s = 0.70$ – 1.00) settings. Across the entire study, there are 4,900 unique page layouts (47,533 unique line instances) yielding 55,130 evaluated line instances across all partitions (Table 1).

Partition	Pages	Lines	Evaluation Role & Content Status	
train	3,500	30,000	Optimization	(unique pages)
val	500	6,334	Checkpointing	(unique pages)
seen_mod	300	3,708	In-distribution baseline	
unseen_mod	300	3,691	Font generalization test	
tiers_clean	300	3,800	Clean control (shared layouts)	
tiers_mod	300	3,800	Moderate control (shared layouts)	
tiers_hard	300	3,797	Hard noise test (shared layouts)	

Table 1: Partition summary. The 3 tier partitions evaluate identical 300 page layouts under 3 degradation settings; total evaluated instances = 55,130.

5. Model Architecture and Experimental Setup

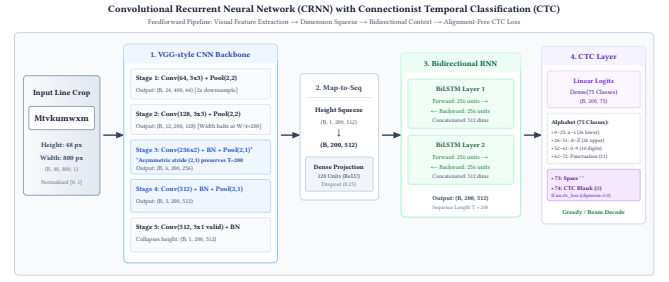


Figure 1: CRNN architecture: 5-stage VGG CNN with asymmetric pooling preserving sequence length at $W/4 = 200$, map-to-sequence squeeze, 2-layer stacked BiLSTM (512 total hidden units), and 75-class CTC output layer.

The CRNN network ingests grayscale line crops ($B, H = 48, W = 800, C = 1$).

1. **CNN Feature Extractor**: Stages 1–2 apply Conv2D (64, 128 filters) with standard 2×2 pooling, reducing dimensions to 12×200 . Stages 3–4 apply asymmetric pooling (2, 1), preserving horizontal width at $W/4 = 200$ while reducing height. Stage 5 applies a valid 3×1 convolution collapsing vertical height to 1.
2. **Map-to-Sequence Squeeze**: The tensor ($B, 1, 200, 512$) is squeezed to ($B, 200, 512$) and projected through a dense layer (128 units, ReLU, 0.25 dropout).
3. **Stacked BiLSTM**: A 2-layer stacked BiLSTM with 256 units per direction (512 concatenated dimensions).
4. **CTC Output Layer**: Linear projection to 75 classes: 73 alphanumeric and punctuation glyphs, whitespace (ID 73), and the CTC blank token $\emptyset$ (ID 74).

CTC loss minimizes negative log-likelihood over all valid alignments:

$$P(Y|X) = \sum_{\pi \in \mathcal{B}^{-1}(Y)} \prod_{t=1}^T P(\pi_t | X), \quad \mathcal{L}_{\text{CTC}} = -\ln P(Y|X)$$

Optimization uses Adam with gradient norm clipping at 5.0.

6. Results

6.1 Training Dynamics & Output Progression

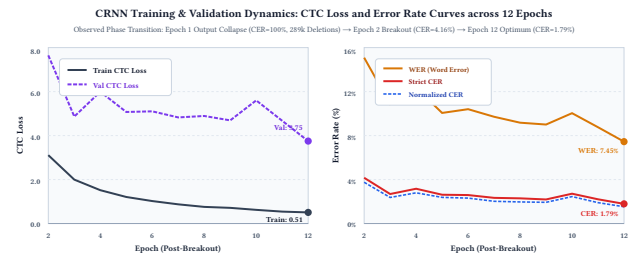


Figure 2: Training dynamics over 12 epochs: (Left) CTC loss curves; (Right) Strict CER (1.79%), Normalized CER (1.53%), and WER (7.45%) convergence.

Inspection of training logs reveals a distinct empirical progression:

- **Epoch 1 (Empty String Output)**: Training loss is 57.33 and validation loss is 630.35. Strict CER is 100% and exact line sequence accuracy is 0.0%, with **289,413 deletions (100% of target characters deleted)** and every prediction emitted as an empty string ("").
- **Plausible Theoretical Interpretation**: Because continuous frame-level softmax activations were not preserved during training, internal activation trajectories cannot be directly confirmed. However, a plausible hypothesis consistent with CTC optimization theory is that near-uniform initial softmax logits make emitting the CTC blank token across all frames a low-entropy local attractor prior to the emergence of coherent temporal alignment gradients. We emphasize that this mechanism represents a plausible explanatory hypothesis rather than a directly verified internal activation measurement.

- **Epoch 2 Breakout:** Optimization overcomes this initial state: Strict CER drops to 4.16%, sequence accuracy rises to 54.58%, and deletion errors decrease from 289,413 to 835.
- **Epoch 12 Convergence:** The checkpoint achieves 1.79% Strict CER, 1.53% Normalized CER, 7.45% WER, and 68.80% exact line accuracy (4,358 of 6,334 lines matched) on the validation set. Training loss reaches 0.51 while validation loss is 3.75, reflecting evaluation on the separate validation font family pool (val/).

6.2 Quantitative Benchmark Results

Partition	Strict CER	Norm CER	WER	Seq. Acc.
Seen Mod.	0.31%	0.20%	1.71%	89.91%
Unseen Mod.	3.99%	3.78%	7.65%	81.60%
Tiers: Clean	0.27%	0.15%	1.43%	90.63%
Tiers: Mod.	0.25%	0.12%	1.34%	91.21%
Tiers: Hard	0.63%	0.47%	3.11%	83.72%

Table 2: Empirical evaluation metrics across the five test conditions.

6.3 Font Shift Evaluation

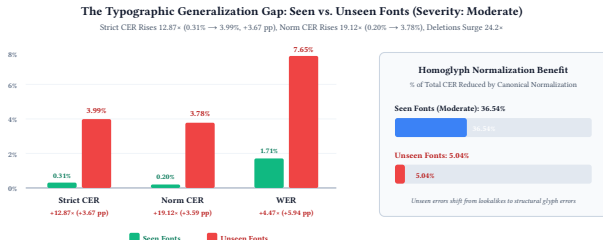


Figure 3: Font shift evaluation: (Left) Metric increases on held-out fonts under moderate noise ($12.87\times$ CER ratio); (Right) Homoglyph normalization recovery rate drops from 36.54% to 5.04%.

Comparing `test_seen_moderate` against `test_unseen_fonts_moderate` under identical moderate degradation reveals:

- **Strict CER:** Increases from 0.00313 (0.31%) to 0.03985 (3.99%), an error ratio of $12.87\times$ (or an increase of **3.67 percentage points**).
- **Normalized CER:** Escalates from 0.00198 to 0.03785, an error ratio of $19.12\times$ (+3.59 percentage points).
- **WER:** Rises from 1.71% to 7.65% (a $4.47\times$ ratio, +5.94 percentage points).
- **Sequence Accuracy:** Decreases by **8.31 percentage points** (from 89.91% to 81.60%), representing a relative decrease of 9.24%.

6.4 Degradation Robustness Across Tiers

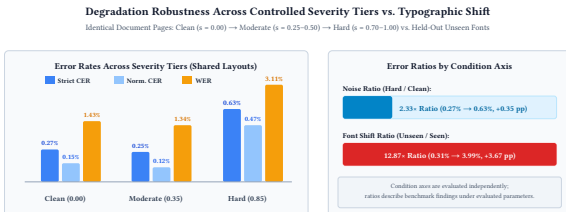


Figure 4: Degradation robustness across controlled tiers on identical page layouts: (Left) Progression from Clean (0.27% CER) to Hard (0.63% CER); (Right) Error ratios by condition axis.

Across the identical page layouts of `test_tiers_*`:

- Moving from Clean ($s = 0.0$) to Hard ($s = 0.70-1.00$) increases Strict CER from 0.27% to 0.63%, representing a **2.33x ratio** (an increase of **0.35 percentage points**), while exact line sequence accuracy drops from 90.63% to 83.72% (a decrease of 6.91 percentage points).
- In this benchmark, the error rate under Hard degradation on seen fonts (0.63% CER) remains substantially lower than that on unseen

fonts under moderate degradation (3.99% CER). We note that font shift and photometric degradation are evaluated along separate axes with specific parameters; this empirical finding describes the evaluated benchmark rather than establishing a universal constant across all possible visual noise models.

6.5 Error Decomposition & Lookalike Normalization

Partition	Substitutions	Deletions	Insertions
Seen Mod.	386 (68.8%)	86 (15.3%)	89 (15.9%)
Unseen Mod.	4,808 (65.1%)	2,082 (28.2%)	494 (6.7%)
Tiers Clean	367 (73.7%)	70 (14.1%)	61 (12.2%)
Tiers Hard	818 (71.3%)	227 (19.8%)	103 (9.0%)

Table 3: Levenshtein error operation counts and proportions.

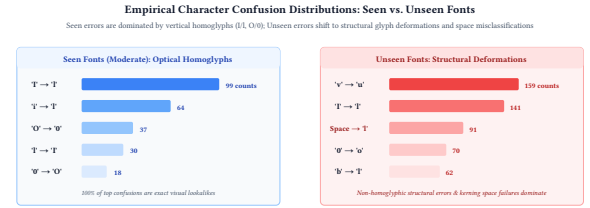


Figure 5: Empirical character confusion breakdown: Seen errors are dominated by vertical homoglyphs ($\frac{l}{i}$, $\frac{o}{0}$); Unseen errors shift to structural deformations ($v \rightarrow u$, $b \rightarrow l$) and kerning space insertions (space $\rightarrow l$).

The lookalike normalization protocol canonicalizes characters into eight equivalence groups: $[I, l, 1] \rightarrow l$, $[O, 0, o] \rightarrow 0$, $[c, C] \rightarrow c$, $[s, S] \rightarrow s$, $[v, V] \rightarrow v$, $[w, W] \rightarrow w$, $[x, X] \rightarrow x$, $[z, Z] \rightarrow z$. Both references and predictions are normalized prior to edit distance evaluation:

$$\text{Norm CER} = \frac{\text{Levenshtein}(\text{norm}(Y), \text{norm}(\hat{Y}))}{|Y|}$$

The percentage reduction is calculated as $\frac{\text{Strict CER} - \text{Norm CER}}{\text{Strict CER}} \times 100\%$.

On seen fonts, normalization reduces CER by **36.54%** (from 0.313% to 0.198%), with errors dominated by vertical bar homoglyphs ($I \rightarrow l$: 99, $i \rightarrow l$: 64, $O \rightarrow 0$: 37). On unseen fonts, normalization reduces CER by only **5.04%** (from 3.985% to 3.785%). Errors shift to structural glyph misclassifications ($v \rightarrow u$: 159, $0 \rightarrow o$: 70, $b \rightarrow l$: 62) and kerning space insertions (space $\rightarrow l$: 91). Furthermore, character deletions exhibit a **24.2-fold explosion** (from 86 to 2,082). A plausible interpretation is that unfamiliar glyph proportions and kerning profiles fail to trigger confident character emissions, resulting in decoded string omissions. Because internal network activations were not logged, this mechanism remains a hypothesis consistent with CTC decoding dynamics rather than a directly measured activation state.

6.6 Character Detection vs. Sequence Transduction

CRNN Line Transduction vs. YOLO11 Spatial Character Detection on Identical Pages
 Methodological Disconnect: High 2D Box Detection mAP (94.56%) vs. Reconstructed OCR Line Transcription (42.86% CER)

YOLO11 Character Detector	CRNN Sequence Recognizer
Full-Page Character Bounding Boxes	ID Rectified Line Strip Transduction
Detection mAP50:	94.56%
Detection mAP50-95:	79.02%
Transcription CER:	42.86%
Word Error Rate (WER):	110.12%
	Seen Moderate CER: 0.31%
	Normalized CER: 0.20%
	Sequence Accuracy: 89.91%
	Word Error Rate (WER): 1.71%

Figure 6: Task formulation comparison under the evaluated pipeline: High 2D detection mAP (94.56%, mAP50-95 of 79.02%) diverges from OCR string transcription CER (42.86%) due to spacing and line-serialization errors.

Partition	mAP50	mAP50-95	YOLO CER	CRNN CER
Seen Mod.	94.56%	79.02%	0.31%	0.20%
Unseen Mod.	94.56%	79.02%	3.99%	3.78%
Tiers Clean	94.56%	79.02%	0.27%	0.15%
Tiers Hard	94.56%	79.02%	0.63%	0.47%

Seen Mod.	94.56%	79.02%	42.86%	0.31%
Unseen Mod.	89.98%	74.15%	46.18%	3.99%
Tiers Clean	94.32%	78.84%	43.79%	0.27%
Tiers Hard	91.99%	76.51%	44.27%	0.63%

Table 4: Comparison between YOLO11 2D detection mAP (mAP50 and mAP50–95) and line transcription CER.

YOLO11 achieves high 2D localization accuracy (mAP50 = 94.56%, mAP50–95 = 79.02%), but its reconstructed text line transcription CER exceeds 42% (WER > 109%). Object detection mAP evaluates isolated bounding-box overlap, whereas CER evaluates sequential string fidelity. In the evaluated reconstruction pipeline, character boxes are grouped into lines and whitespace is inferred via geometric gap thresholding. Because whitespace has no physical bounding box, slight threshold misalignments corrupt word tokenization, compounding Levenshtein distance.

Under the evaluated implementation, high character-detection accuracy did not translate into comparably accurate reconstructed text transcription. This result illustrates the distinction between character localization and sequence transcription; it does not establish that all YOLO-based OCR systems are inferior to CRNN-based systems.

7. Discussion

The empirical results show that the evaluated CRNN architecture maintains strong transcription accuracy under moderate synthetic photometric distortions (blur, noise, skew), but exhibits substantial vulnerability to typographic shifts on held-out font families. Under unseen fonts, character deletions surge 24.2-fold (from 86 to 2,082). While internal activation arrays were not logged to verify feature maps directly, this deletion surge is hypothesized to reflect alignment failures where unfamiliar glyph geometries do not trigger label emission during CTC decoding.

Furthermore, the comparison with YOLO11 highlights the distinction between spatial detection and linguistic transcription. While object detectors localize glyph instances accurately, integrated sequence transduction with an intrinsic whitespace token remains better aligned with end-to-end text line transcription.

8. Threats to Validity and Limitations

- 1. Synthetic Domain Shift:** Evaluations use synthetic images from synth_pages. Generalization to physical scanned documents remains unmeasured.
- 2. Lack of Language Model Prior:** Text lines were sampled uniformly at random without lexical n-gram priors. This isolates visual sequence recognition, but prevents semantic disambiguation.
- 3. Confusion Analysis Scope:** Available evaluation logs record the top-5 confusion pairs per split. Hypothesized pairs outside the top-5 threshold could not be evaluated.
- 4. Absence of Line-Length Partitioning:** Error metrics are aggregated across line crops without stratifying by sequence length.
- 5. Unavailable Internal Activations:** Continuous frame-level softmax activations and hidden recurrent states were not logged during training or inference. Consequently, proposed explanations regarding blank-token attractors and alignment failure modes remain plausible hypotheses grounded in CTC optimization theory and observed output strings, rather than directly confirmed internal activation measurements.

9. Conclusion and Future Work

This study evaluated the generalization limits of a CRNN-CTC model trained on synthetic document lines. On seen fonts under moderate degradation, the model achieved 0.31% Strict CER. When evaluated on held-out font families, Strict CER rose to 3.99% (a $12.87\times$ ratio, +3.67 percentage points), accompanied by a 24.2-fold increase in deletions. Across identical layouts, photometric noise increased CER by a $2.33\times$

ratio (+0.35 percentage points). Possible future research directions include evaluating broader training font diversity, integrating language-model-constrained beam search, and testing zero-shot transfer on historical document scans.

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Appendix: Reproducibility Details

- **Input Processing:** Grayscale [0, 255] inverted and normalized to [0, 1], zero-padded to (48, 800, 1).
- **Optimizer & Schedule:** Adam ($\beta_1 = 0.9, \beta_2 = 0.999, \epsilon = 10^{-8}$), learning rate 1×10^{-3} , 12 epochs.
- **Gradient Clipping:** Norm 5.0. Batch size 32.
- **Receptive Field:** Horizontal stride 4 across CNN stages yielding $T = 200$ feature slices.
- **Vocabulary Alphabet:** 73 ASCII characters + Space (ID 73) + CTC Blank (ID 74).